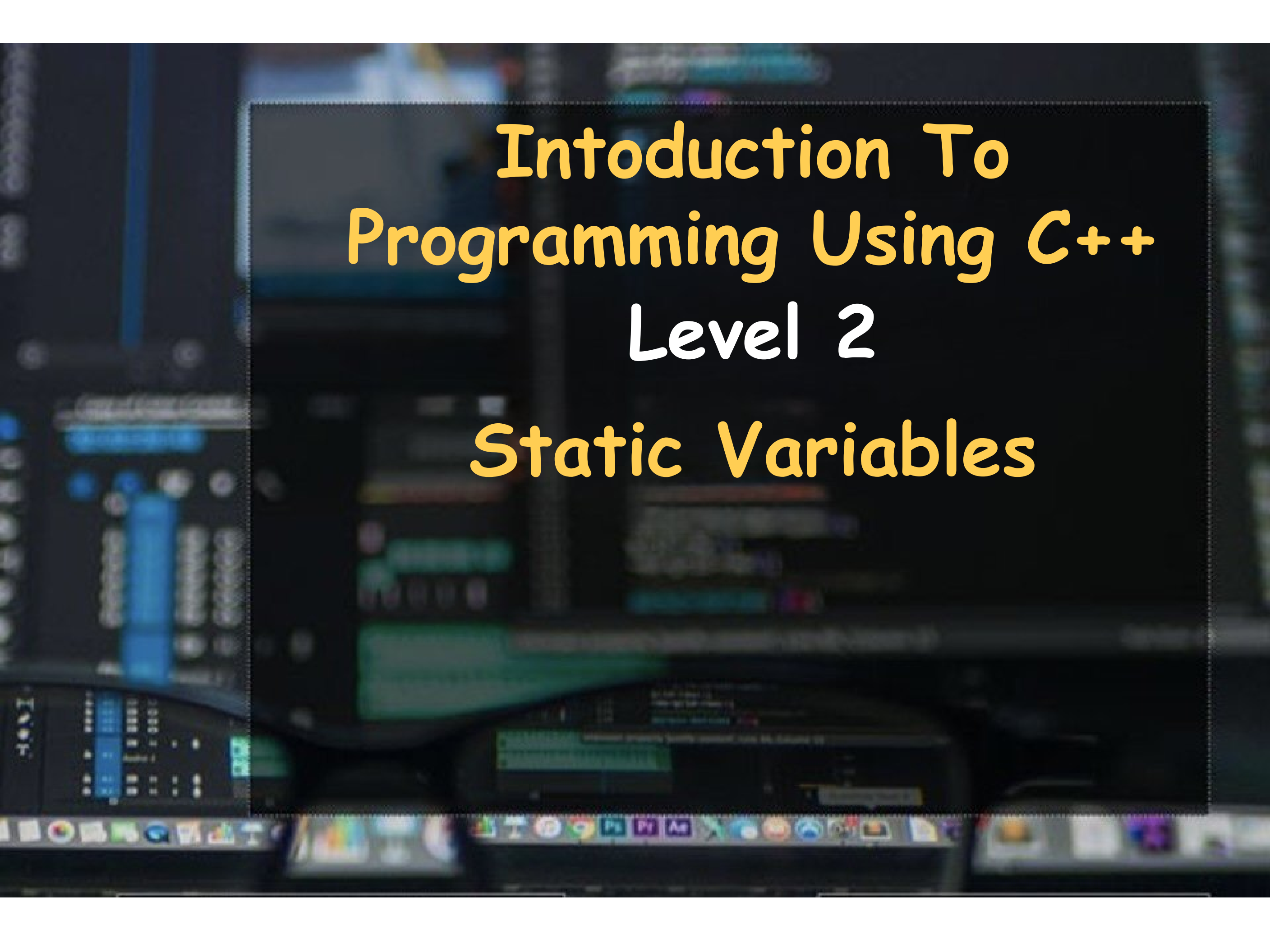
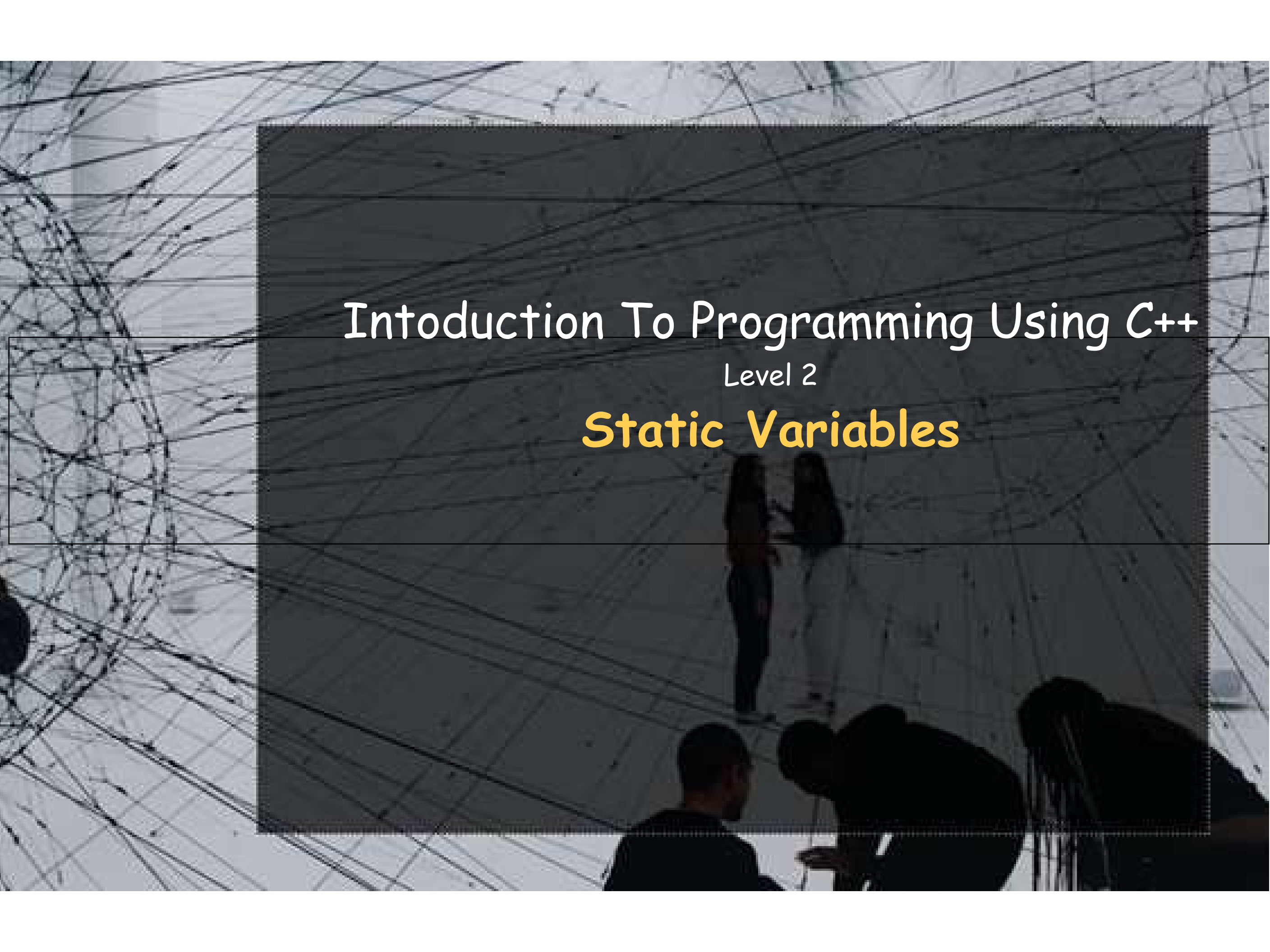




Introduction To Programming Using C++ Level 2 Static Variables



Introduction To Programming Using C++

Level 2

Static Variables

Definition

■ A Static Variable is a variable that keeps its value throughout the lifetime of the program.

■ A local `static` variable has Local Scope, but its lifetime is extended until the program ends.

▨ Unlike a normal Local Variable, a Static Variable is not created and initialized again every time the Function is called.

Explanation

- ◆ When a normal Local Variable is declared inside a Function, it is created when the Function is called and its lifetime ends when the Function finishes.
- ◆ Therefore, every new Function Call starts with a new Local Variable and its initial value.
- ◆ When the `static` keyword is used, the variable keeps its value between Function Calls.
- ◆ The variable is initialized only once, and later Function Calls continue using the same stored value.

💡 Example

◆ Normal Local Variable:

```
void MyFunc()
{
    int Number = 1;
    cout << "Value of Number: " << Number << "\n";
    Number++;
}
int main()
{
    MyFunc();
    MyFunc();
    MyFunc();
    return 0;
}
```

➡ Result

```
Value of Number: 1
Value of Number: 1
Value of Number: 1
```

Note

- ◆ Now we add the `static` keyword:
- ◆ The complete example becomes:

```
void MyFunc()
{
    static int Number = 1;
    cout << "Value of Number: " << Number << "\n";
    Number++;
}
int main()
{
    MyFunc();
    MyFunc();
    MyFunc();
    return 0;
}
```

Result

```
Value of Number: 1
Value of Number: 2
Value of Number: 3
```

⊘ Warning

- ◆ A Static Variable can make code harder to understand if its changing value is not obvious.
- ◆ Avoid using Static Variables when the same behavior can be handled more clearly by passing a value through Parameters.

Comparison

Local Variable

- ✓ Has Local Scope inside the Function.
- ✓ Created when the Function is called.
- ✓ Its lifetime ends when the Function finishes.
- ✓ Reinitialized each time the Function is called.
- ✓ Does not keep its value between Function Calls.

Static Local Variable

- ✓ Has Local Scope inside the Function.
- ✓ Initialized only once.
- ✓ Keeps its value between Function Calls.
- ✓ Remains in memory until the program ends.
- ✓ Useful when a Function needs to remember information from previous Calls.

✦✦ Summary

- ✓ `static` allows a Local Variable to keep its value between Function Calls.
- ✓ A Static Variable is initialized only once.
- ✓ Its Scope remains Local to the Function.
- ✓ Its lifetime continues until the program ends.
- ✓ A normal Local Variable is reinitialized on every Function Call.
- ✓ Static Variables are useful when a Function needs to remember a previous state.

Thank you 😊